

Dylan Joseph

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Education

University of North Texas Denton, TX
Computer Science Track; GPA: 4.0/4.0 2025 – Expected 2027

- Expected to complete 60+ accredited university credits by 2027; UNT President's List, Fall 2025 and Spring 2026.

Research Experience

University of North Texas Denton, TX
Computer Vision Researcher 2025 – Present

- Develop AI approaches for first-person multi-object tracking and image reconstruction for optical Doppler tomography medical-imaging datasets.
- Contributed annotations to approximately 7% of VQS-4K and received a \$4,000 scholarship for a 10-week project using 2.1 million frames.

UT Dallas Computer Vision and Multimodal Computing Lab Richardson, TX
Computer Vision Research Intern Summer 2025

- Extended Sound of Pixels with early visual-feature fusion for visually conditioned audio generation.
- Developed a UCF-101 transformer for action recognition and implemented CNN, custom ResNet, and transfer-learning ResNet-18 models.

Projects

AudiLex AI | Voice-Based Law Study 2025 – Present

- Develop the frontend and application features for a voice-based study platform for Northwestern University law students. Support textbook retrieval, voice interaction, and deployment.

ScamMap | Python, Deepgram, Real-Time Audio | [GitHub](#) 2026

- Built a scam-detection application that transcribes microphone audio, identifies speakers, and displays risk assessments with supporting evidence.
- Ran the application on one CPU with live risk checks and repeat-speaker matching; won second place and a \$500 prize at the Finovate Hackathon.

Selected Deep Learning Portfolio | PyTorch, Transformers, CNNs 2025 – 2026

- UCF-101 Transformer: Developed a two-layer, eight-head transformer with a 512-dimensional embedding that reached approximately 77% action-recognition accuracy.
- Modified LoCoNet: Adapted LoCoNet for AVDIAR active-speaker detection and reached 88% average accuracy.
- KMNIST DC-GAN: Built a deep convolutional generative adversarial network in PyTorch to generate Kuzushiji characters from KMNIST.
- Audio-Visual Early Fusion: Developed a 3D-ResNet early-fusion model for sound separation and localization.

UNT Grades | TypeScript, Data Visualization | [untgrades.app](#) 2026 – Present

- Built and maintain a UNT course grade-distribution viewer that served 533 visitors and generated 2,813 page views.

oh-my-codex – Upstream Contributor | TypeScript | [Commits](#) 2026

- Contributed two commits merged into upstream `main` for a TypeScript project that coordinates Codex CLI agents.
- Prevented stale worktrees from blocking later team runs and reduced an agent prompt by 33% while preserving behavior and passing full CI.

Try Omarchy – Upstream Contributor | Linux Guest Setup | [Merged PR #195](#) 2026

- Added a U.S. International keyboard layout with dead key accents to guest onboarding for ANSI keyboards, including a pinned backport and verification tests.

UNT VSB – Contributor | Development, Deployment, Performance, Backend | [Live Site](#) 2026

- Contributed across application development, deployment, performance, and backend engineering for UNT VSB.

Technical Skills

Languages: Python, Java, C++